

Code

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.model_selection import train_test_split
4 from sklearn.ensemble import RandomForestClassifier
5 from sklearn.metrics import accuracy_score, classification_report
6
7 # Load Dataset
8 df = pd.read_csv('/content/Iris (1).csv')
9
10 # Features and Target
11 X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
12 y = df['Species']
13
14 # Split Data
15 X_train, X_test, y_train, y_test = train_test_split(
16     X, y, test_size=0.2, random_state=42
17 )
18
19 # Train Model
20 model = RandomForestClassifier(random_state=42)
21 model.fit(X_train, y_train)
22
23 # Prediction
24 y_pred = model.predict(X_test)
25
26 # Accuracy
27 print("Accuracy:", accuracy_score(y_test, y_pred))
28
29 # Classification Report
30 print(classification_report(y_test, y_pred))
31
32 # Visualization
33 plt.figure(figsize=(6,4))
34 for species in df['Species'].unique():
35     data = df[df['Species'] == species]
36     plt.scatter(
37         data['SepalLengthCm'],
38         data['SepalWidthCm'],
39         label=species
40     )
41 plt.xlabel("Sepal Length")
42 plt.ylabel("Sepal Width")
43 plt.title("Iris Flower Classification")
44 plt.legend()
45 plt.show()
```

Output

Accuracy: 1.0

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Plot

